Agent Architecture Overview

System Architecture

```
graph TB
    subgraph CONFIG["📋 Project Configuration"]
        CLAUDE_MD["CLAUDE.md<br/>Architecture, constants, packages"]
        STANDARDS["STANDARDS.md<br/>Unity 6 APIs, conventions, CI"]
        AGENTS_MD["AGENTS.md<br/>Beads workflow, session protocol"]
        ROUTING[".claude/agents/routing.json<br/>Label-based dispatch rules"]
    end

    subgraph INFRA["⚙️ Infrastructure"]
        RUNNER["agent-runner.ps1<br/>PowerShell polling loop (30s)<br/>Launches claude -p
per task"]
        BD["bd (beads CLI)<br/>Dolt-backed issue tracker<br/>Labels, deps, sync"]
        GIT["Git + GitHub<br/>Code persistence<br/>bd sync hooks"]
    end

    subgraph MEMORY["🧠 Agent Memory"]
        HANDOFF_SCHEMA["handoffs/SCHEMA.md<br/>GUPP-Lite v1 format"]
        ACTIVITY["handoffs/activity.jsonl<br/>Append-only event log"]
        AGENT_HANDOFFS["handoffs/<agent>.json<br/>Last session state per agent"]
        AGENT_LOCAL[".claude/agents/<name>/<br/>handoffs/<name>.json<br/>Per-agent local
handoff copy"]
        SETTINGS[".claude/agents/<name>/<br/>.claude/settings.local.json<br/>Per-agent
permissions"]
    end

    RUNNER -->|"polls bd ready<br/>--label agent:X"| BD
    BD -->|"work found"| RUNNER
    RUNNER -->|"launches claude -p<br/>with task prompt"| AGENTS
    BD <-->|"bd sync"| GIT
    CONFIG -.->|"read on session start"| AGENTS
    AGENT_HANDOFFS -.->|"context recovery"| AGENTS
    AGENTS -.->|"write on session end"| AGENT_HANDOFFS
    AGENTS -.->|"append events"| ACTIVITY

    subgraph AGENTS["🤖 Agent Pool (9 Agents)"]
        direction TB

        subgraph CORE_AGENTS["Core Gameplay"]
            PLAYER["🎮 player<br/>──────────────<br/>Movement, input,<br/>abilities, dash,
<br/>double jump, powerups<br/>──────────────<br/>⏳ b8z: Consolidate<br/>input assets"]
            COMBAT_ENEMY["👾 enemy-behavior<br/>──────────────<br/>AI state machines,
<br/>bosses, sensors,<br/>wave spawning<br/>──────────────<br/>🔒 6bd: Boss encounters<br/>
🔒 80c: Zone grouping"]
            SYSTEMS_AGENT["🔧 systems<br/>──────────────<br/>GameManager, Save,<br/>Health,
Mana, XP,<br/>Wind, Cutscenes<br/>──────────────<br/>⏳ mn2: Inventory<br/>⏳ bxi: Balance
curves"]
```

```
        end

        subgraph WORLD_AGENTS["World & Presentation"]
            ENVIRONMENT["🏰 environment<br/>──────────────<br/>Tilemaps, platforms,
<br/>hazards, doors,<br/>levers, interactables<br/>──────────────<br/>🔒 r7p: Exploration
gates"]
            CAMERA["📷 camera<br/>──────────────<br/>Follow, parallax,<br/>bounds, boss
rooms,<br/>shake, zoom<br/>──────────────<br/>No active beads"]
            VFX_AGENT["✨ vfx<br/>──────────────<br/>Particles, fog,<br/>precipitation,
<br/>hit/death effects<br/>──────────────<br/>No active beads"]
            SOUND["🔊 sound-design<br/>──────────────<br/>SFXManager, Music,<br/>UISoundBank,
<br/>audio pipeline<br/>──────────────<br/>No active beads"]
        end

        subgraph META_AGENTS["Meta & Interface"]
            ARCHITECT["📐 architect<br/>──────────────<br/>Patterns, CI, reviews,<br/>tech
debt, system<br/>design, refactoring<br/>──────────────<br/>⏳ 9ag: Prefab audit<br/>⏳ i17:
Zone system<br/>⏳ 17p: Unit tests<br/>⏳ buu: CI pipeline"]
            UI_UX["🎨 ui-ux<br/>──────────────<br/>Gothic UI, menus,<br/>HUD, skill trees,
<br/>accessibility<br/>──────────────<br/>🔒 4za: World map"]
        end
    end

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    style INFRA fill:#1a1a2e,stroke:#0f3460,color:#eee
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    style META_AGENTS fill:#16213e,stroke:#533483,color:#eee
```

Current Blocking Chain

```
graph LR
    subgraph COMPLETED["✅ Completed"]
        SO["1s2: ScriptableObject<br/>data assets<br/><b>P0 ✓ CLOSED</b>"]
    end

    subgraph BOTTLENECK["🚧 Critical Path (In Progress)"]
        PREFABS["9ag: Audit prefabs<br/><b>P0 ⏳ architect</b>"]
        ZONES["i17: Multi-scene<br/>zone system<br/><b>P1 ⏳ architect</b>"]
    end

    subgraph BLOCKED["🔒 Blocked (6 beads)"]
        CI["1j4: Prefab CI check<br/>P2"]
        BOSSES["6bd: Boss encounters<br/>P1"]
        ENEMIES["80c: Zone enemy groups<br/>P1"]
        EXPLORE["r7p: Exploration gates<br/>P1"]
        NPC["7hq: NPC/dialogue<br/>P3"]
        MAP["4za: World map UI<br/>P3"]
```

```
    end

    SO -->|"unblocked"| ZONES
    PREFABS --> CI
    ZONES --> BOSSES
    ZONES --> ENEMIES
    ZONES --> EXPLORE
    ZONES --> NPC
    ZONES --> MAP

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    style BOTTLENECK fill:#7f4f24,stroke:#dda15e,color:#eee
    style BLOCKED fill:#6a040f,stroke:#d00000,color:#eee
```

Agent Session Lifecycle

```
sequenceDiagram
    participant R as agent-runner.ps1
    participant BD as bd (beads)
    participant C as claude -p
    participant G as Git/GitHub

    loop Every 30s
        R->>BD: bd ready --label agent:X --json --limit 1
        BD-->>R: [] (no work) or [{issue}]
    end

    Note over R,BD: Work found!

    R->>C: Launch with task prompt
    activate C

    C->>C: Read STANDARDS.md + CLAUDE.md
    C->>C: Read handoffs/<agent>.json (prior context)
    C->>BD: bd update <id> --claim
    C->>BD: bd show <id>

    Note over C: RPI: Research → Plan → Implement

    C->>C: Write/edit code files
    C->>BD: bd close <id> --reason "summary"
    C->>C: Write handoffs/<agent>.json
    C->>C: Append to activity.jsonl
    C->>G: git add + commit + pull --rebase + push
    C->>BD: bd sync

    deactivate C
    C-->>R: Exit (code 0)
    R->>R: Cooldown 10s, resume polling
```

Project Stats Snapshot

Metric	Count
Total beads	62
Closed	49
In Progress	7
Open (unblocked)	0
Blocked	6
Ready to work	0
Agents with handoffs	3 (vfx, systems, sound-design)

Legend: ⏳ In Progress | 🔒 Blocked | ✅ Completed